SmartTrade

Project Name

SmartTrade – Vendor, Customer & Product Catalogue Marketplace

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1. Project Objective

SmartTrade is a mobile application designed to digitize the product catalogue, dealer discovery, quotation, invoice and local business ecosystem for industries such as:

- Electrical
- Plumbing
- Cement
- Iron & Steel
- Hardware
- Paints
- Construction materials
- Other related products

The application connects:

1. **Personal Users / Customers**
2. **Business Users / Dealers / Shops**
3. **Companies / Manufacturers / Brands**
4. **Service Providers**
5. **Job Seekers**
6. **Employers**

The primary objective is to make it easier for customers to find products and genuine/verified dealers, while helping businesses create professional quotations and invoices using standardized company product catalogues.

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2. User Registration Types

SmartTrade will support the following major user types:

2.1 Personal User / Customer

Free registration.

2.2 Business User / Dealer / Shop

Registration with a minimal fee.

2.3 Company / Manufacturer / Brand

Company registration for uploading and maintaining official product catalogues. Registration with nominal fee

2.4 Service Provider

Electricians, plumbers, carpenters, technicians, mechanics and other local service providers. Registration with minimal fee.

2.5 Job Seeker

Users looking for local employment. Registration with minimal fee

2.6 Employer

Businesses/vendors looking to hire local employees.

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3. Personal User / Customer

3.1 Purpose

The main purpose of Personal Users is to help customers easily discover products, company catalogues and dealers in their local area.

Personal Users should be able to view available product catalogues according to their requirements.

Example

A customer wants to purchase a **Havells electrical wire**.

The customer should be able to:

1. Search for Havells wire.
2. View the available Havells products.
3. View product images.
4. View product name and model/code.
5. View product description/specifications.
6. View the applicable catalogue/listed price.
7. Identify dealers selling that product.
8. Search dealers using:
    - PIN code
    - Area/locality name
    - District name
9. View dealer/shop details.
10. Contact the dealer.

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4. Personal User Registration

4.1 Registration Type

Free Registration

4.2 Required Fields

- Full Name
- Mobile Number
- OTP Verification
- Occupation / Profession
- PIN Code
- Area / Locality
- District
- State
- Location

4.3 Validation

- Mobile number must be unique.

- OTP verification is mandatory.
- Duplicate registration should not be allowed.

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5. Personal User Features

Personal Users can:

Product Discovery

- Browse product catalogues.
- Search by product name.
- Search by product code/model.
- Search by brand.
- Browse category-wise products.
- View product images.
- View company catalogue information.
- View product descriptions/specifications.
- View applicable prices.

Dealer Discovery

Customers can search for dealers based on:

- PIN code
- Area/locality
- District
- Location

Example:

Search: Havells Wire

System displays:

- Product
- Brand
- Available dealers
- Shop/Business name
- Dealer location
- Area
- PIN code
- Distance

- Contact number
- Dealer verification status
- Authorized dealer status, where applicable

If the product is unavailable in the selected location:

> "Product not available in the selected location."

The application may suggest dealers in nearby locations.

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6. Business User / Dealer / Shop

6.1 Purpose

The main purpose of the Business User module is to digitize the quotation and invoicing process for dealers and shops.

Currently, a dealer may sell many different types of products from different companies.

For example, an electrical shop may sell:

- Havells
- Legrand
- Polycab
- V-Guard
- Anchor
- Other electrical brands

A plumbing/hardware shop may sell products from many different manufacturers.

Because of this, preparing quotations manually can be difficult because the dealer needs to:

- Find different product catalogues.
- Find product codes.
- Find product descriptions.
- Check prices.
- Calculate discounts.
- Find HSN codes.
- Prepare quotation documents.
- Prepare invoices.

SmartTrade will digitize this process.

The Business User can use the available company catalogues to quickly create professional quotations and invoices.

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7. Business User Registration

Business Users will register with a **minimal registration fee**.

7.1 Required Business Information

- Business Name
- Owner/Authorized Person Name
- GST Number
- Trade License / Firm Registration / Applicable License Number
- Mobile Number
- Email Address
- Business Address
- PIN Code
- Area / Locality
- District
- State
- Business Category

Business Categories

Examples:

- Electrical
- Plumbing
- Cement
- Iron & Steel
- Hardware
- Paints
- Construction Materials
- Other

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8. Business Category & Brand Selection

During registration, the Business User must specify the types of products and brands they sell.

Example

If the dealer selects:

Category: Electrical

The system displays available brands:

- Havells
- Legrand
- Polycab
- V-Guard
- Anchor
- Schneider
- Finolex
- Other

The dealer can select multiple brands.

Example

Electrical

☑ Havells
☑ Legrand
☑ Polycab
☐ V-Guard
☐ Anchor
☑ Other

If the required brand is not available:

Other Brand

The user must select:

Other

and enter the brand name.

Example:

Brand Name: ABC Electricals

This information should be sent to the Admin for review.

Purpose

This information will help SmartTrade administrators identify new brands being sold by dealers and add those brands to the platform after verification.

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9. Business User Trade Worker Information

During Business User registration, the dealer/shop must provide details of at least **10 people** associated with their business network.

Examples:

- Electricians
- Plumbers
- Carpenters
- Painters
- HVAC technicians
- Other trade workers

Required Information

For each person:

- Name
- Mobile Number
- Trade / Profession
- Optional area/location

Minimum Requirement

The Business User must provide **at least 10 trade-worker contacts** during registration.

Example:

Name	Mobile Number	Profession

Person 1	XXXXXXXX01	Electrician
Person 2	XXXXXXXX02	Electrician
Person 3	XXXXXXXX03	Plumber
...	...	...
Person 10	XXXXXXXX10	Plumber

This information can later support the SmartTrade service-provider ecosystem.

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10. Business User Validation

The system should validate:

- Mobile Number
- OTP
- GST Number
- Trade/Firm License
- Business information
- Duplicate business registration

The Admin should have the ability to approve/reject/suspend a Business User.

Business verification status should be maintained, for example:

- Pending Verification
- Verified Business
- Rejected
- Suspended

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11. Business User Main Features

Business Users can:

1. View company catalogues.
2. Search products.

3. Search by brand.
4. Search by product code.
5. Add products to cart.
6. Select quantity.
7. Apply discount.
8. Create quotations.
9. Share quotations with customers.
10. Convert approved quotations into invoices.
11. Generate PDF quotations.
12. Generate PDF invoices.
13. Maintain product availability.
14. Maintain stock quantity.
15. Maintain selling price.
16. View their selected product categories and brands.

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12. Product Catalogue

The Product Catalogue is the central module of SmartTrade.

Catalogues should be available to:

- Personal Users
- Business Users

Company/Manufacturer users will upload and maintain the catalogues.

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13. Catalogue Structure

The catalogue hierarchy should be:

Category

↓

Company / Brand

↓

Product

↓

Product Model / Variant

↓

Product Details

↓

Product Image

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Catalogue PDF / Company Document

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14. Product Information

Each product should support the following information:

- Category
- Company / Brand
- Product Name
- Product Code
- Model Number
- Product Description
- Product Specifications
- Product Image
- Unit
- Catalogue/List Price
- HSN Code
- GST Rate
- Catalogue PDF/reference
- Product Status
- Availability

The system should maintain a unique product code/model wherever possible to reduce duplicate products.

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15. Product Categories

Electrical

Examples:

- Havells
- V-Guard
- Anchor
- Polycab
- Legrand
- Other approved brands

Each brand can contain multiple products/models and prices.

Plumbing

Examples:

- Ashirvad
- Astral
- Vimal
- Other approved brands

Cement

Examples:

- UltraTech
- ACC
- Other approved brands

Iron & Steel

Examples:

- TMT Bars
- Structural Steel
- Other products

Additional categories can be added by Admin.

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16. New User Type – Company / Manufacturer / Brand

16.1 Purpose

Companies/Manufacturers/Brands will have a dedicated registration type.

The main purpose is to allow companies to upload and maintain their official product catalogues directly on SmartTrade.

Why Companies Should Upload Catalogues

Currently companies may need to:

- Print product catalogues.
- Send catalogues to dealers.
- Distribute catalogues throughout India.
- Update/reprint catalogues when product information or prices change.

SmartTrade can reduce this dependency by providing a centralized digital catalogue.

Companies can upload their official catalogue once and make it available digitally to registered users and dealers.

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17. Company User Registration

Required Fields

- Company Name
- Brand Name
- Authorized Person Name
- Mobile Number
- Email
- GST Number
- Company Registration Details

- Registered Address
- PIN Code
- Area
- District
- State
- Website, if applicable
- Company Logo
- Company Documents
- Contact Information

Validation

- Mobile OTP verification
- Email verification
- GST verification where applicable
- Company/document verification by Admin
- Duplicate company/brand check

Company status:

- Pending Verification
- Verified
- Rejected
- Suspended

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18. Company Catalogue Management

Verified Company Users can:

- Upload catalogues.
- Add new products.
- Update existing products.
- Add product images.
- Add product codes.
- Add model numbers.
- Add product descriptions.
- Add specifications.
- Add HSN codes.
- Add GST information.
- Update catalogue/list prices.
- Upload catalogue PDFs.

- Deactivate discontinued products.
- Add new product models/variants.

All catalogue changes should maintain version/update information.

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19. Company Dealer Mapping

One of the major advantages of Company registration is **dealer mapping**.

Companies can provide/manage their authorized dealer information.

The system should allow mapping:

Company

↓

Brand

↓

Product

↓

Authorized Dealer

↓

Dealer Location

↓

PIN Code / Area / District

This can help customers identify dealers and can reduce duplicate or incorrect dealer/product information.

Where a company officially confirms a dealer, the dealer should display:

Company Authorized Dealer

The system should distinguish this from a normal verified business.

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20. Product Duplicate Prevention

The system should try to prevent duplicate products.

Products should preferably be identified using:

- Company/Brand
- Product Code
- Model Number
- Product Name
- Variant

If a company has already uploaded a product, another user should not create an identical product record unnecessarily.

Admin should have the ability to review potential duplicate products.

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21. Product Availability Mapping

Business Users can maintain:

- Product Availability
- Stock Quantity
- Selling Price
- Availability Status

Example:

Havells Wire – Product Code XXXXX

Dealer A:

- Available
- Stock: 25
- Price: ₹XXXX

Dealer B:

- Available

- Stock: 10
- Price: ₹XXXX

Personal Users can then find dealers based on the selected location.

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22. Vendor/Dealer Search Module

Personal User searches:

- Product
- Brand
- PIN Code
- Area
- District

Example:

Search: Havells Switches

System displays:

- Product
- Available dealers
- Shop Name
- Location
- PIN Code
- Distance
- Contact Number
- Verification Status
- Authorized Dealer Status, if confirmed

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23. Cart Module

Business Users can select products from company catalogues and add them to a cart.

Cart should contain:

- Product Image
- Product Name

- Product Code
- Product Description
- Brand
- Quantity
- Catalogue/List Price
- Discount
- HSN Code
- GST
- Final Price
- Total

The cart can be converted into a quotation.

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24. Quotation Module

Business User Flow

Search Catalogue

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Select Products

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Add to Cart

↓

Enter Quantity

↓

Apply Discount

↓

Review Cart

↓

Generate Quotation

↓

Share with Customer

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25. Quotation Details

Quotation should contain:

- Business/Dealer Name
- Business Address
- GST Number
- Customer Name
- Customer Contact
- Quotation Number
- Quotation Date
- Product Image
- Brand
- Product Name
- Product Code
- Product Description
- HSN Code
- Quantity
- Catalogue/List Price
- Discount %
- Discount Amount
- Taxable Amount
- GST
- Final Price
- Total Amount
- Terms & Conditions

Quotation should be downloadable/shared as a PDF.

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26. Invoice Module

Invoice Flow

Step 1:

Business User creates quotation

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Step 2:

Quotation is shared with Customer

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Step 3:

Customer approves quotation

↓

Step 4:

Business User converts quotation into Invoice

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Step 5:

Invoice generated and shared as PDF

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27. Invoice Details

Invoice should contain:

- Invoice Number
- Invoice Date
- Business Name
- GST Number
- Business Address
- Customer Details
- Product Image
- Brand
- Product Name
- Product Code
- Product Description

- HSN Code
- Quantity
- Price
- Discount
- Taxable Amount
- GST
- Total Amount
- Payment Status
- Terms & Conditions

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28. Service Provider Module

Service Providers may include:

- Electricians
- Plumbers
- Carpenters
- Painters
- Technicians
- Mechanics
- HVAC technicians
- Other skilled workers

Registration fields:

- Full Name
- Mobile Number
- OTP
- Profession/Trade
- PIN Code
- Area
- District
- State
- Location

Validation:

- Mobile number unique
- OTP mandatory
- Duplicate registration prevention

Personal Users can search for service providers based on:

- Trade
- PIN Code
- Area
- District
- Location

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29. Job Seeker Module

Job Seekers can:

- Create profile.
- Add skills.
- Add experience.
- Select preferred location.
- Search local jobs.
- Apply for jobs.
- Receive notifications.

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30. Employer Module

Employers/Business Users can:

- Create job openings.
- Specify required skills.
- Specify salary/range.
- Specify location.
- Review applicants.
- Contact suitable candidates.
- Manage recruitment status.

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31. Admin Panel

Admin will manage:

Users

- Personal Users
- Business Users
- Company Users
- Service Providers
- Job Seekers
- Employers

Business Verification

- GST verification
- License verification
- Business approval
- Business suspension

Company Verification

- Company verification
- Brand verification
- Catalogue approval
- Product approval

Product Catalogue

- Categories
- Brands
- Products
- Product models
- Product images
- Catalogue PDFs
- Product codes
- HSN codes
- Duplicate product review

Dealer Management

- Verified dealers
- Authorized dealer mapping
- Dealer locations
- Product availability

Transactions

- Quotations
- Invoices

Reports

- Total Personal Users
- Total Business Users
- Total Companies
- Total Dealers
- Total Service Providers
- Product Reports
- Catalogue Reports
- Quotation Reports
- Invoice Reports
- Category Reports
- Brand Reports
- Dealer Reports

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32. Recommended Application Modules

The application should contain at least the following modules:

1. User Management
2. Personal User Management
3. Business User Management
4. Company/Manufacturer Management
5. Service Provider Management
6. Job Seeker Management
7. Employer Management
8. Product Catalogue Management
9. Brand Management
10. Product Management
11. Dealer Management
12. Dealer Search
13. Product Availability
14. Cart
15. Quotation Management
16. Invoice Management
17. Company Dealer Mapping
18. Notifications
19. Admin Panel

20. Reports & Analytics

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33. Notification Module

Notifications may include:

- OTP
- Registration approval
- Business verification
- Company verification
- New catalogue/product
- Product availability
- Quotation received
- Quotation approved
- Invoice generated
- Job notifications
- Service requests

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34. Business Model

Personal Users

Free Registration

Personal Users can browse catalogues and find dealers/service providers without registration fees.

Business Users

Minimal Registration Fee

Business Users receive additional functionality such as:

- Catalogue access
- Product selection
- Cart
- Quotation generation

- Invoice generation
- Business profile
- Product availability
- Dealer/shop presence
- Customer quotation management

Company Users

Company/Manufacturer registration can be offered as a company/business service.

Potential future commercial options:

- Catalogue hosting
- Brand profile
- Product promotion
- Dealer management
- Dealer mapping
- Analytics
- Premium company features

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35. Important Product & Dealer Verification Rules

SmartTrade should maintain separate verification statuses.

Business Verification

Verified Business

Means SmartTrade has verified the business information/documents.

Company Verification

Verified Company / Brand

Means the company/brand has been verified by SmartTrade.

Authorized Dealer

Company Authorized Dealer

Should only be displayed when the company has confirmed/mapped that business as an authorized dealer.

These statuses should not be treated as the same.

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36. Technical Requirements

Mobile Application

- Android
- iOS

Backend

- REST API
- Authentication Service
- OTP Service
- User Management
- Product Catalogue Service
- Quotation Service
- Invoice Service
- Notification Service

Database

- PostgreSQL / MySQL

Cloud

- AWS / Azure

Storage

Storage for:

- Company catalogues
- Product images
- Company documents

- Business documents
- PDF quotations
- PDF invoices

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37. High-Level Application Flow

Personal User

Personal User Registration

↓

Search Product

↓

View Company Catalogue

↓

View Product Details

↓

Search Dealer

↓

Select PIN Code / Area / District

↓

View Available Dealers

↓

Contact Dealer / Purchase

Business User

Business Registration

↓

Select Business Categories

↓

Select Brands

↓

Add Other Brands if Required

↓

Provide Trade Worker Information

↓

Business Verification

↓

Browse Company Catalogues

↓

Select Products

↓

Add to Cart

↓

Create Quotation

↓

Send to Customer

↓

Customer Approval

↓

Generate Invoice

Company User

Company Registration

↓

Company Verification

↓

Create/Upload Brand Catalogue

↓

Add Products

↓

Add Product Images & Details

↓

Publish Catalogue

↓

Map Authorized Dealers

↓

Customers/Dealers View Catalogue

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38. Overall SmartTrade Ecosystem

Company / Manufacturer

↓

Official Product Catalogue

↓

Verified Dealers / Shops

↓

Business User creates Quotation

↓

Customer receives Quotation

↓

Customer approves

↓

Invoice Generated

↓

Product Purchased

At the same time:

Personal User

↓

Search Product

↓

Search PIN Code / Area / District

↓

Find Dealer

↓

Contact Dealer

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39. Future Enhancements – Phase 2

- Online Payments
- WhatsApp Quotation Sharing
- Vendor Ratings
- Company/Dealer Ratings
- Order Tracking
- Advanced Stock Management
- Purchase Orders
- Delivery Tracking
- Multi-language Support
    - English
    - Telugu
    - Hindi
- AI Product Recommendations
- Customer Order History
- Dealer Order History

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40. Future Enhancements – Phase 3

- ERP Integration
- GST Portal Integration
- Advanced Inventory Management
- Analytics Dashboard
- Company Sales Analytics
- Dealer Performance Analytics
- Product Demand Analytics
- Automated Dealer Management
- API integration with company systems

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41. Key Business Value

SmartTrade is intended to solve three major problems.

For Customers

Customers can easily find:

- Products
- Product information
- Company catalogues
- Nearby dealers
- Verified businesses
- Authorized dealers where company confirmation exists

For Dealers / Business Users

Dealers can easily:

- Find products
- Use company catalogues
- Create quotations
- Apply discounts
- Add HSN/GST information
- Generate professional invoices
- Reduce manual quotation preparation

For Companies / Manufacturers

Companies can:

- Digitally distribute catalogues.
- Reduce catalogue printing costs.
- Reduce physical catalogue distribution.
- Maintain updated product information.
- Publish official product information.
- Manage dealer mapping.
- Help customers find authorized dealers.
- Reduce duplicate/incorrect product records.

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42. Core Objective

The core objective of SmartTrade is to create a digital ecosystem connecting **Companies, Dealers, Customers and Service Providers**.

Companies provide official product catalogue information.

Dealers use those catalogues to prepare quotations and invoices.

Customers use the application to discover products and locate dealers.

This creates a structured digital marketplace for electrical, plumbing, cement, iron, hardware, construction and related products while reducing manual catalogue and quotation processes.